MILITARY SPECIFICATION

TEST SET, TOW TARGET, A/E24T-143

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. The equipment covered by this specification shall provide for electrical continuity test as follows:

a. Electrical wiring provision for installation of the A/A47U-3 or A/A47U-4 Tow Target Reeling Machine Launcher System on U.S. Navy aircraft and A/A24U-21 Tow Target Control Set on A-6E aircraft.

b Bench test of the JEK-131/A24U-21 Interconnecting Box.

1.2 General The complete A/E24T-143 Tow Target Control Test Set shall consist of a light and switch panel test unit, a removable reeling machine launcher control simulator and a set of electrical test cables. The complete test set shall be housed in a suitable portable enclosure for environmental protection.

1.3 Classification.

1.3.1 Type. The test set specified herein shall meet the requirements of Type I equipment as defined in MIL-T-28800.

1.3.2 Class The test set specified herein shall meet the requirements of Class 2 equipment as defined in MIL-T-28800 for type of enclosure and for environmental operating parameters.

1.3.2.1 Enclosure. The enclosure for the test set shall meet the requirements of Style T enclosure as defined in MIL-T-28800.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (Code 9313), Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

DOD-D-1000	Drawings, Engineering and Data Lists
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-28800	Test Equipment for Use with Electrical and Electronic Equipment, General Specification for

STANDARDS

Military

MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-480	Configuration Control Drawing Changes, Deviations and Waivers

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NADC-TE22140	Schematic, Reel/Launcher Control Simulator
NADC-TE22141	Electrical Schematic, Tow Target Test Set A/E24T-143

(Application for copies should be addressed to: Naval Air Development Center (Code 6012), Warminster, PA 18974.)

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2.1.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. In the event of conflict between the drawings and figures references or included herein, and the performance requirements of this specification the performance requirements shall take precedence over the data provided.

3. REQUIREMENTS

3.1 First article. When specified, a sample test set shall be subjected to first article inspection (see 4.4, 6.2 and 6.3).

3.2 Safety The test set shall incorporate the safety features required by MIL-T-28800 to safeguard personnel and equipment during connection, operation, maintenance, repair, and interchanging of a complete equipment, assembly, or component thereof.

3.2.1 Personnel hazard protection Personnel hazard protection shall be provided in accordance with MIL-T-28800 as applicable to the test set described herein.

3.3 Parts, materials, and processes

3.3.1 Restricted materials Restrictions on the use of materials invoked by MIL-T-28800 shall apply

3.3.2 Selection of parts, materials, and processes. Selection of parts, materials, and processes shall be in accordance with the requirements for Type I of MIL-T-28800, except that Appendix A of MIL-E-5400 shall be used in lieu of MIL-BULLETIN-400 for applicable specifications and standards. The test unit housing shall be aluminum or high-impact plastic which shall meet requirements for protection against the hazards encountered in a salt air environment so as to be corrosion-resistant and impervious to moisture formation inside the housing.

3.3.3 Finishes Finishes shall conform to the requirements of Type I as specified in MIL-T-28800. The color of the test unit, including the test panels, shall conform to color U as specified in MIL-T-28800. The color of the enclosure case shall conform to color V as specified in MIL-T-28800.

3.4 Design and construction

3.4.1 The design and construction of the test set shall be in accordance with the requirements for Type I of MIL-T-28800, and as detailed and modified herein.

3.4.1.1 Test panels. The test unit shall incorporate a basic test panel and two (2) overlay panels. Suggested layouts of switch, lamp, and connector locations for the basic and overlay panels are shown in Figures 1A thru 1C. Table I provides information for specific usage of the basic and overlay panels.

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TABLE I. Panel usage.

Panel	Use for
Basic (Figure 1A)	JEK-131/A24U-21 Bench Test
Overlay A (Figure 1B)	F-4, A-4 Aircraft
Overlay B (Figure 1C)	A-6E Aircraft Only

3.4.1.2 Shielded construction. Interconnecting leads involving circuits considered susceptible to radiated interference or capable of radiating interference shall be shielded so as to meet the test requirements of MIL-STD-461.

3.4.1.3 Cables, connectors, and cable assemblies. In addition to meeting all requirements for connection and construction, the cable sets which form parts of the test set described herein shall be designed in accordance with Figures 2 and 3.

3.4.1.4 Reel launcher control simulator. The Reel Launcher Control Simulator shall conform to the configuration and dimensions shown in Figure 4.

3.4.1.5 Electrical schematics. Electrical schematics for the A/E24T-143 Test Set and the Reel Launcher Control Simulator shall be in accordance with NADC drawings TE22140 and TE22141 respectively.

3.4.1.6 Self-test feature. Provision shall be made in the design of the test set for a self-test feature.

3.5 Electrical power sources and connections.

3.5.1 Unless otherwise required in the detail schematic(s), the test set shall meet the primary input power conditions of 28 VDC, operate within the required performance limits, and operate with connections as specified for Type I of MIL-T-28800.

3.5.2 Maximum amperage. The test set shall operate at a maximum current of 5 amperes.

3.5.3 Power and connecting cables. Power and connecting wires and cables necessary for the operation of the test set to meet the conditions imposed by this specification shall be a part of the test unit, but shall be removable by separation of the connections when the test set is stored and not in use.

3.6 Dimensions, weight, and mechanical stability.

3.6.1 Dimensions. Dimensions of the test unit shall be proposed by the contractor and subject to approval by the procuring activity.

3.6.2 Weight. The maximum weight of the complete test set shall be proposed by the contractor and subject to approval by the procuring activity.

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3.6.3 Mechanical stability. Mechanical stability shall conform to the requirements of MIL-T-28800.

3.7 Enclosure A portable enclosure for storage of the test unit, reeling machine launcher control simulator, power and interconnection wires and cables, operating sheets or manuals, up to six (6) overlay panels, and accessories as required, shall be provided. The enclosure construction style and finish color shall be as specified herein.

3.8 Marking and identification. Plate markings, panel markings, external enclosure markings, schematic diagrams and marking for items other than plates and panels shall be in accordance with the requirements for Type I of MIL-T-28800 and as detailed herein.

3.8.1 Panel usage and information Information identifying the basic and overlay panels and the respective usage data shown in Table I shall be marked in a prominent location on the container lid.

3.9 Environmental requirements. The test set and its protective storage enclosure shall meet the environmental requirements for Class 2, Style T specified in MIL-T-28800. Applicable environmental tests and test conditions for the class and style specified herein shall be in accordance with MIL-T-28800, except as modified herein.

3.9.1 Warm-up. Warm-up requirements of MIL-T-28800 do not apply for this test set.

3.9.2 Electromagnetic compatibility. Appropriate shielding and filtering shall be employed in the engineering design of the test set so that EMI requirements specified in MIL-T-28800 for Class 2 equipment are met.

3.10 Maintainability The test set shall meet the requirements for Type I maintenance provisions as specified in MIL-T-28800. The requirement for Type I washability shall not apply

3.11 Interchangeability of parts The interchangeability of parts of all test sets built to this specification shall meet the requirements of MIL-T-28800 for Type I equipment.

3.12 Documentation.

3.12.1 Engineering drawings. A full design disclosure set of engineering drawings shall be developed by the contractor to the requirements of DOD-D-1000, Level 3. The following types of engineering drawings shall be developed and maintained under configuration management in accordance with MIL-STD-480.

a. Detail fabrication drawings

b. Assembly drawings

c. Wire lists

d. Schematics

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3.12.2 First article demonstration test report. The contractor shall furnish to the procuring activity a test report indicating results of the tests performed under the first article demonstration requirements of this specification.

3.12.3 Operating and maintenance manual. The contractor shall furnish, as part of each complete test set, a suitable operation and maintenance manual.

3.13 Workmanship. Workmanship shall be in full accordance with the requirements of MIL-T-28800.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-T-28800 except that altitude is not applicable and power input shall be 28 $\pm$ 2 VDC.

4.4 First article inspection First article inspection shall be performed at a laboratory acceptable to the Government on a sample unit produced with equipment and procedures normally used in production.

4.4.1 First article demonstration sample. The first article demonstration sample shall consist of one complete A/E24T-143 Test Set. The sample shall be identical in design and performance to the items which will be produced on the manufacturer's production line. The acceptance test for the first article sample shall consist of successful demonstration and completion of the first article tests specified in 4.4.2.

4.4.2 First article demonstration tests. First article demonstration tests shall consist of the following:

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4.4.2.1 Examinations and tests defined under Groups A, C, D, and E of MIL-T-28800 except that requirements in MIL-T-28800 identified as altitude, fungus, washability, accoustic noise, x-radiation, and peculiar requirements are not applicable.

4.4.2.2 Upon satisfactory completion and approval of tests specified in 4.4.2.1, an operational demonstration verifying form, fit, and function of the test set on A-4, A-6E, and F-4 aircraft and bench test on the JEK-131/A24U-21 Interconnecting Box shall be performed by the procuring activity with representatives of the contractor in attendance.

4.5 Quality conformance inspection. The contractor shall submit to the procuring activity for approval, a proposed quality conformance inspection and test procedure to be used during manufacture of production quantities of the test set.

4.6 Cause for rejection. Failure of a test set to meet any of the requirements specified herein shall be cause for rejection.

4.7 Rejection and retest. Test sets which have been rejected may be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejecting and the action taken to correct the defects found in the original test shall be furnished to the procuring activity

5. PACKAGING

5.1 Preservation, packaging, packing, and marking shall conform to the requirements of MIL-T-28800 (see 6.2.1).

6. NOTES

6.1 Intended use. The intended use of the test set covered by this specification is to provide for electrical continuity testing for:

a. Electrical wiring provisions for installation of the A/A47U-3 or A/A47U-4 Tow Target Reeling Machine Launcher System on U.S. Navy aircraft and A/A24U-21 Tow Target Control Set on A-6E aircraft located on an aircraft carrier or at shore based facilities.

b. Bench testing of the JEK-131/A24U-21 Interconnecting Box.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number and date of this specification.

b. First article inspection sample (4.4.1).

c. First article inspection sample size (4.4.1).

d. First article inspection data or test reports.

e. Levels of preservation, packaging, and packing (5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No	Data requirements	Applicable DID No.	Options
3.12.1	Engineering drawings	DI-E-7031	---
3.12.2	First article test reports	UDI-T-21349	---
3.12.3	Operation and maintenance manuals	DI-M-5105B	---
4.5	Quality conformance inspection and test procedure		

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction sample consisting of one complete A/E24T-143 Test Set as described herein. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article and test reports.

Preparing Activity:
Navy - AS
Project No. 6625-N797

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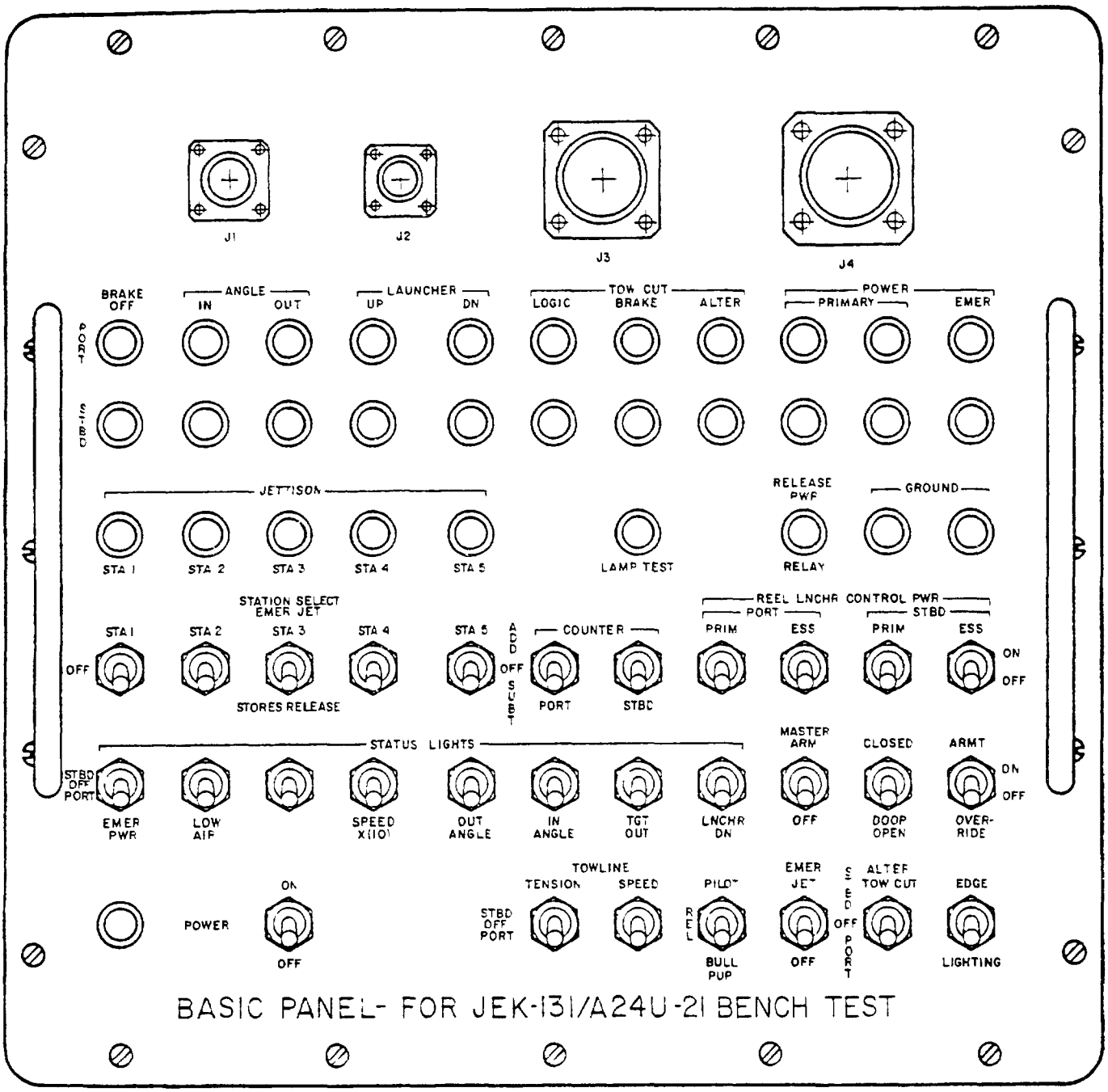


FIGURE 1A. SUGGESTED LAYOUT - BASIC TEST PANEL.
(JEK-131/A24U-21 BENCH TEST)

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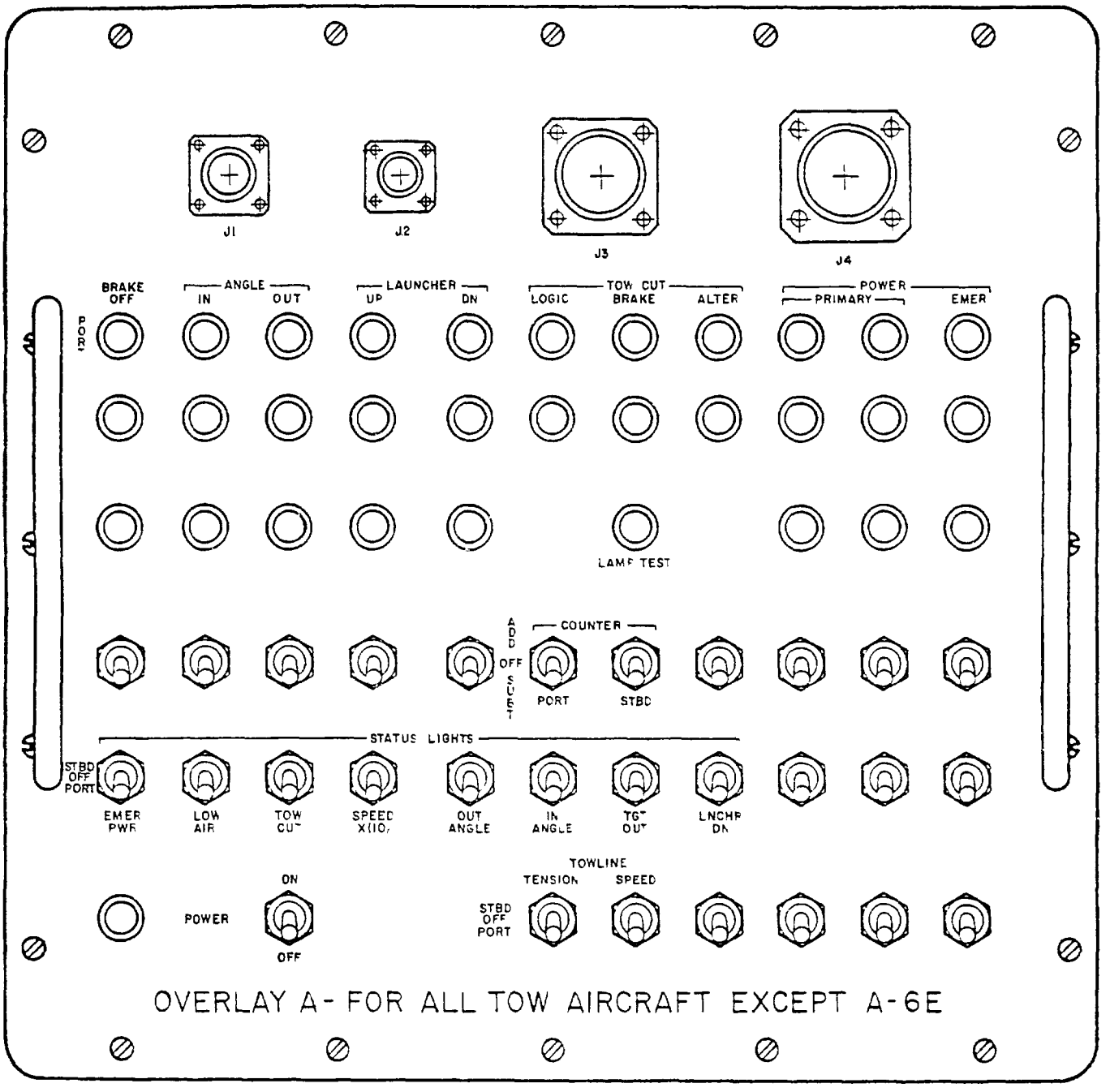


FIGURE 1B. SUGGESTED LAYOUT - OVERLAY TEST PANEL. (A/A47U-3 AND A/A47U-4 ELECTRICAL PROVISIONS TEST ON ALL TOW AIRCRAFT EXCEPT A-6E AIRCRAFT)

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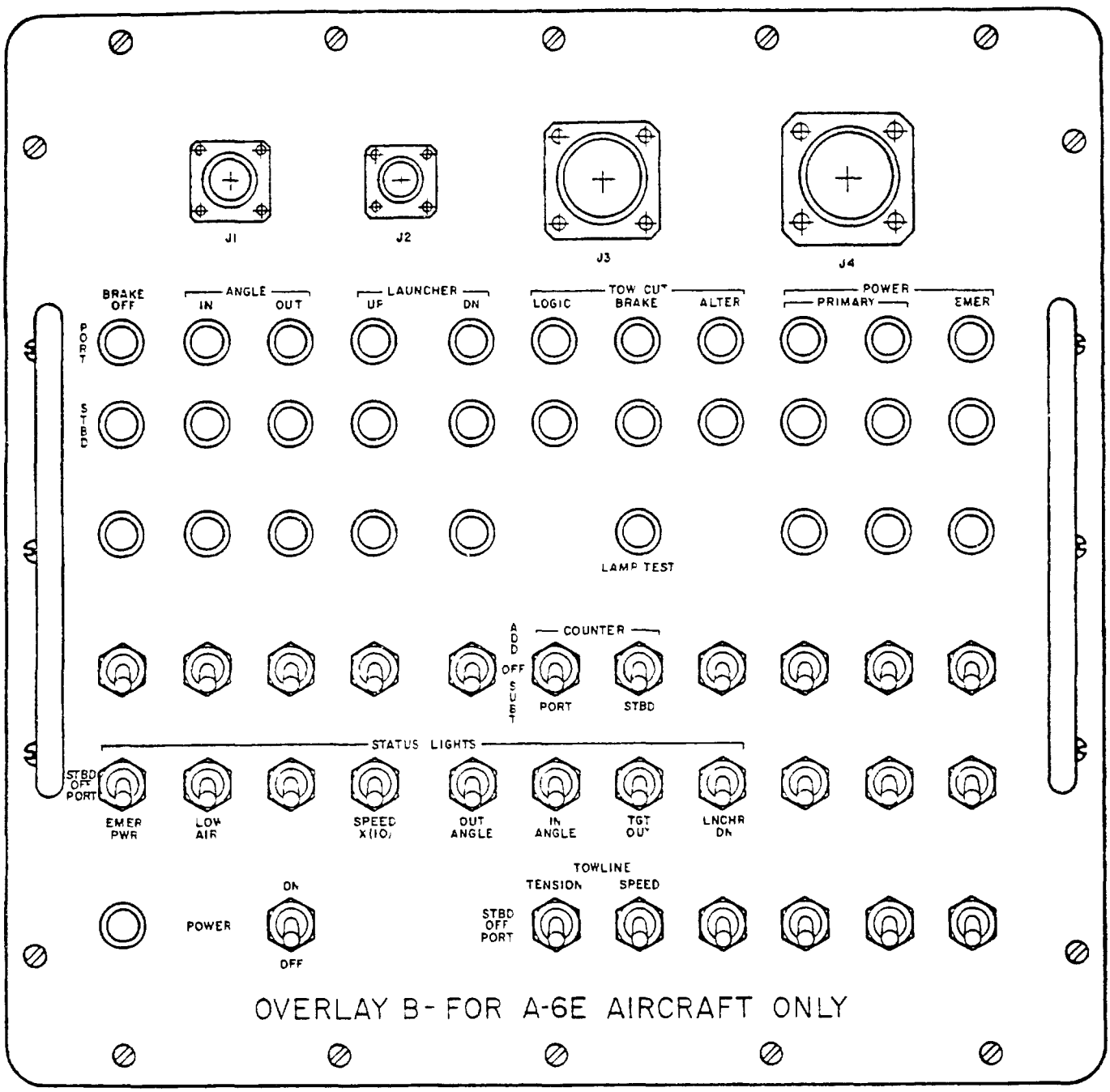
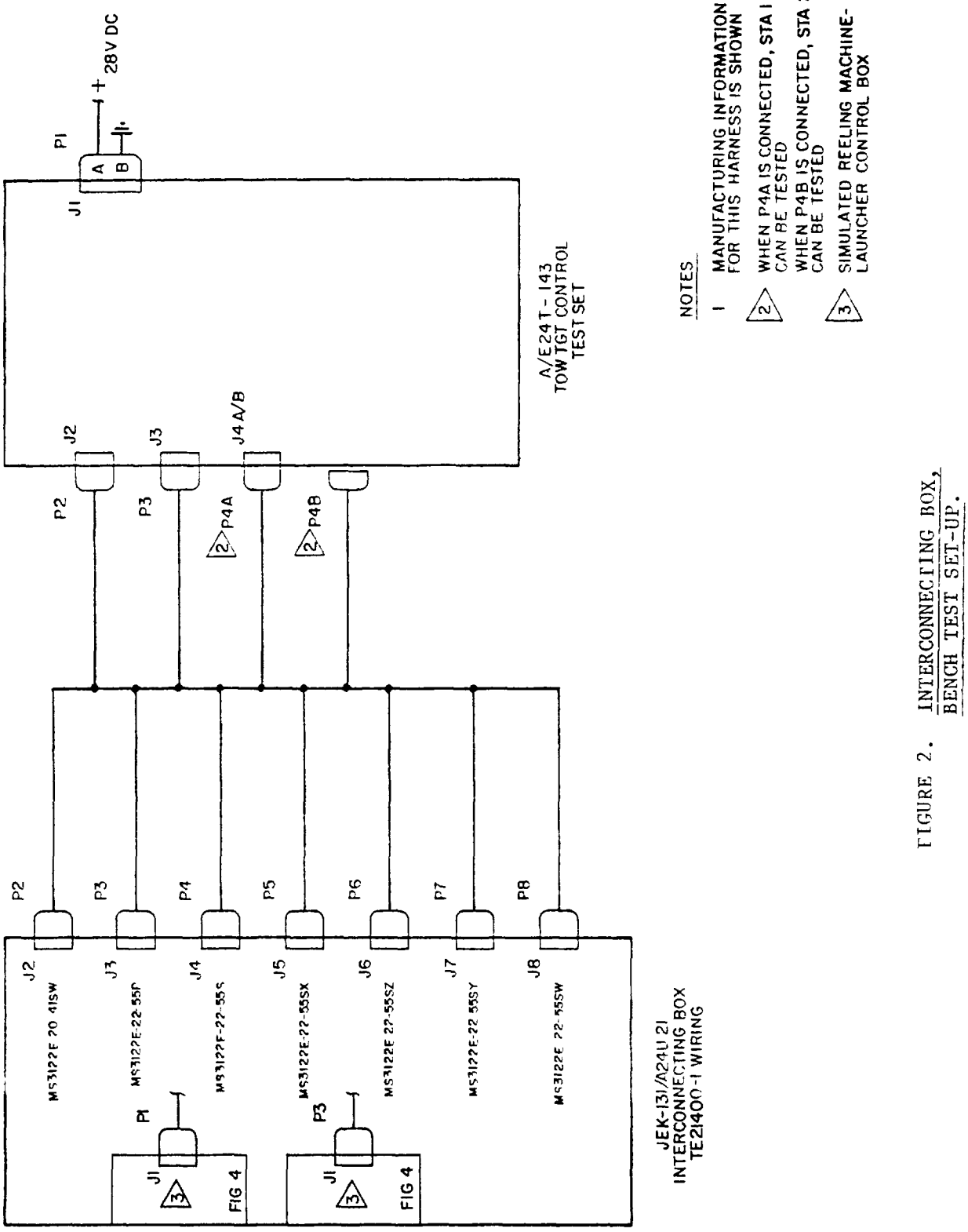


FIGURE 1C. SUGGESTED LAYOUT - OVERLAY TEST PANEL.
(JEK-131/A24U-21 ELECTRICAL PROVISIONS
TEST ON A-6E AIRCRAFT ONLY)

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NOTES

1 MANUFACTURING INFORMATION FOR THIS HARNESS IS SHOWN

2 WHEN P4A IS CONNECTED, STA 185 CAN BE TESTED
WHEN P4B IS CONNECTED, STA 284 CAN BE TESTED

3 SIMULATED REELING MACHINE-LAUNCHER CONTROL BOX

FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP.

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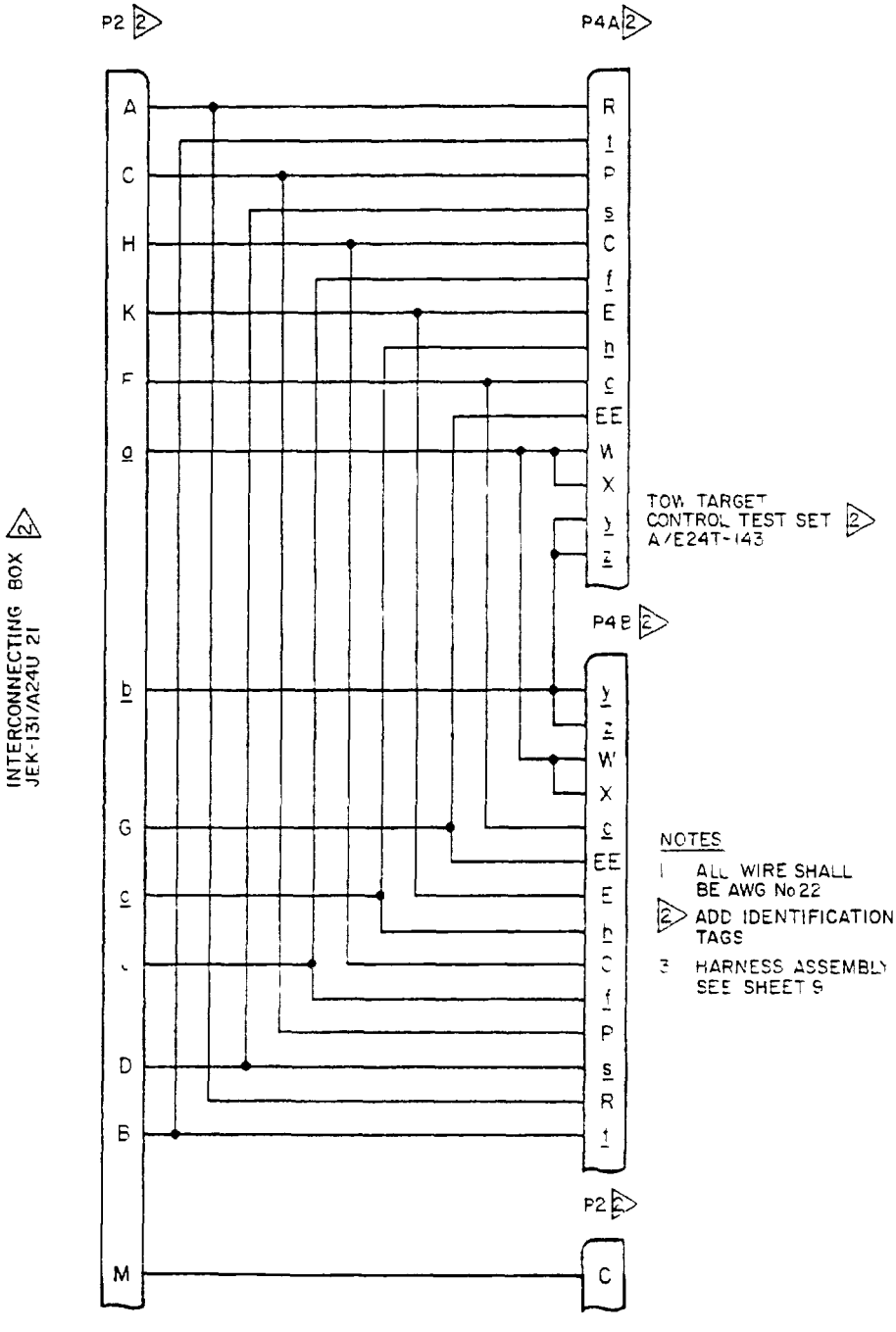


FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)

SHEET 2 OF 9

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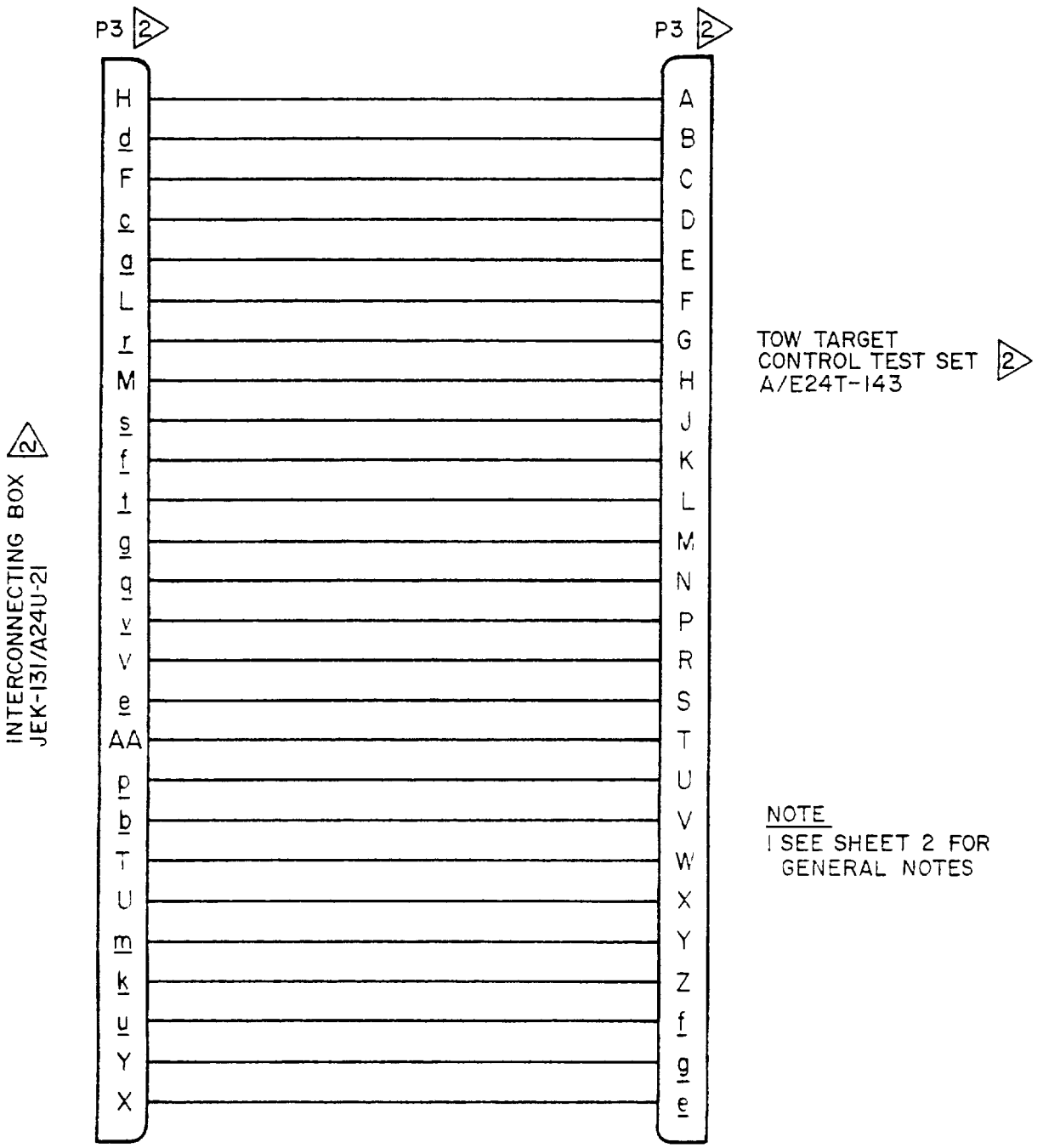


FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)

SHEET 3 OF 9

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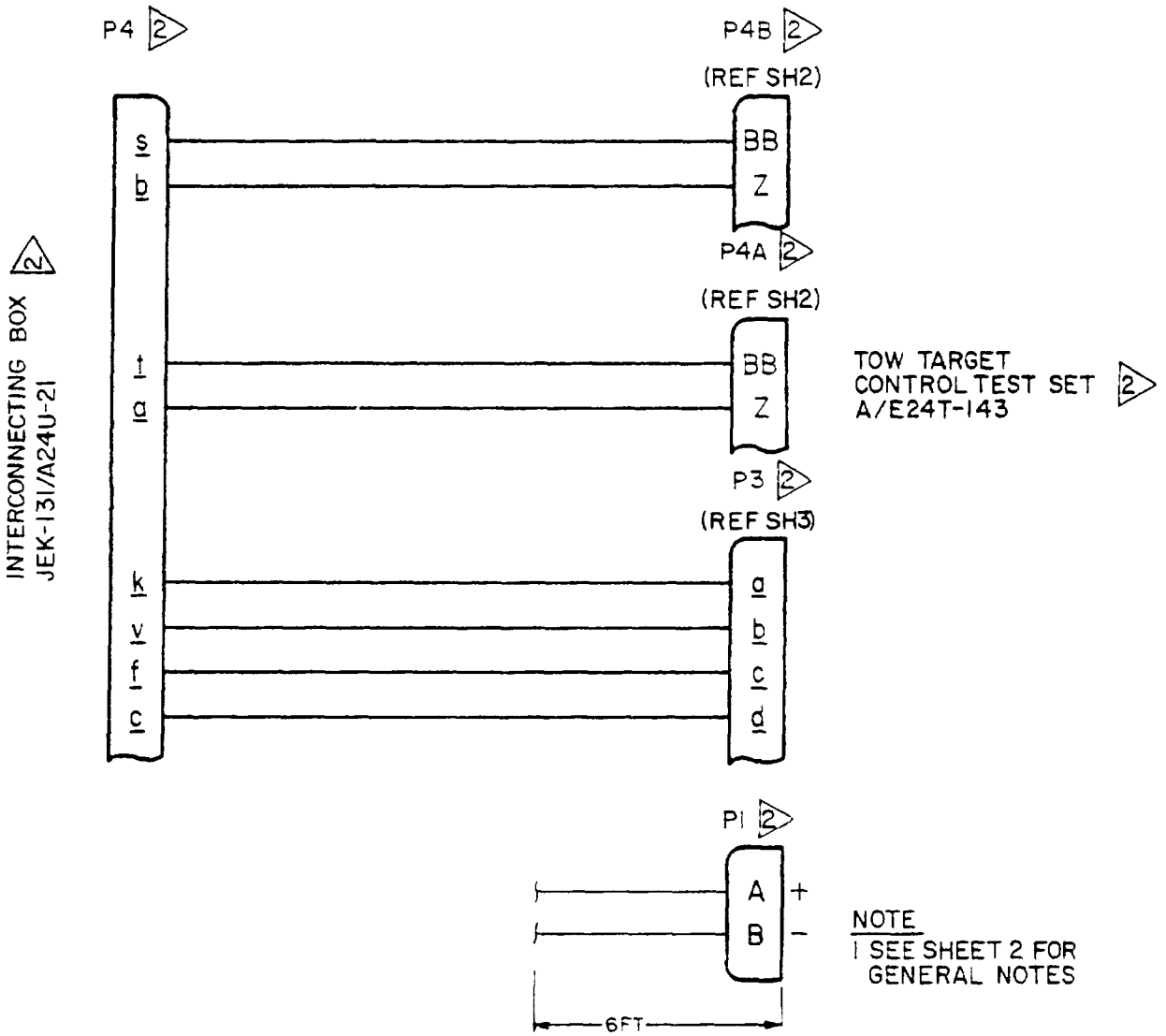
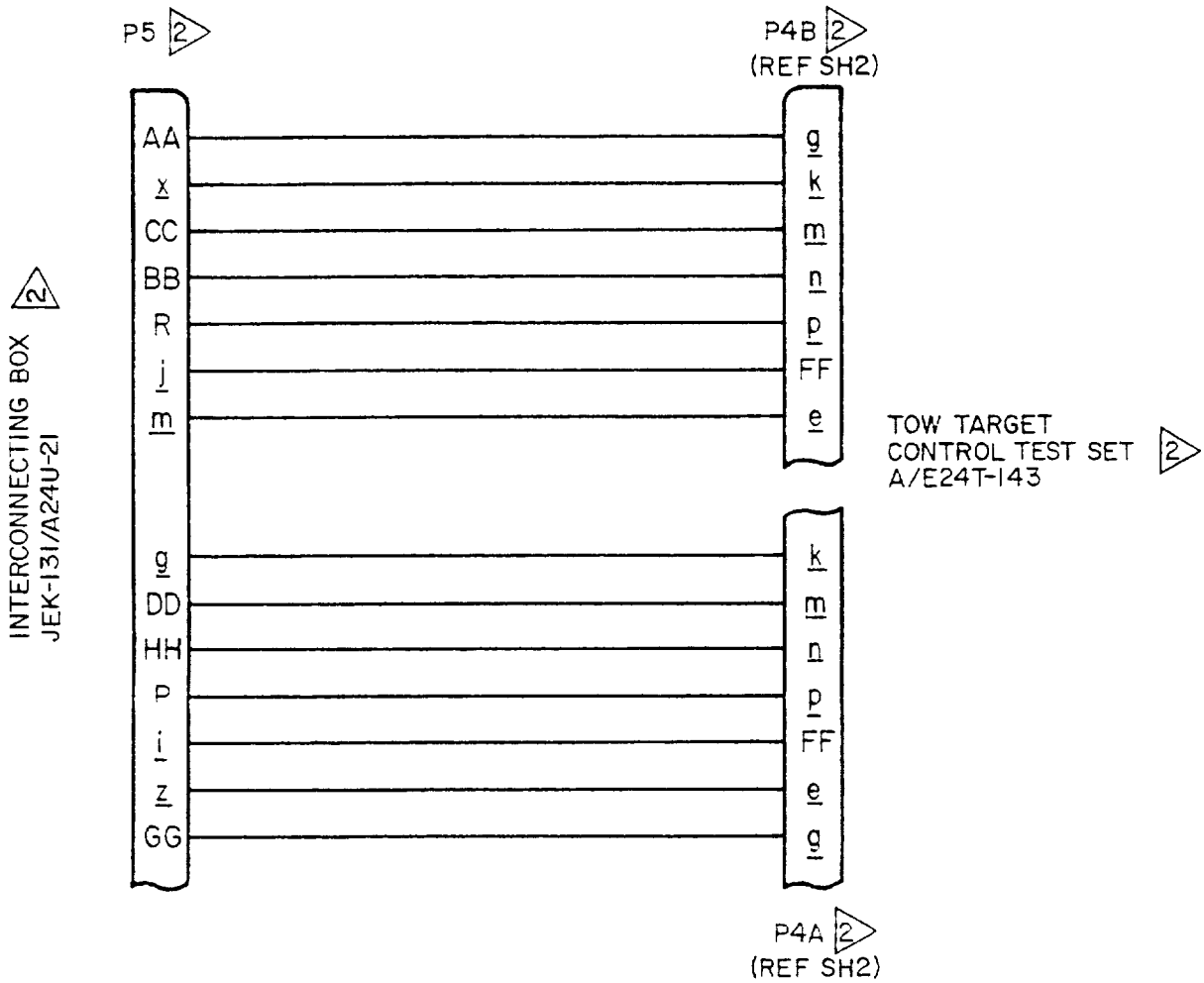


FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)

SHEET 4 OF 9

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NOTE
1 SEE SHEET 2 FOR GENERAL NOTES

FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)

SHEET 5 OF 9

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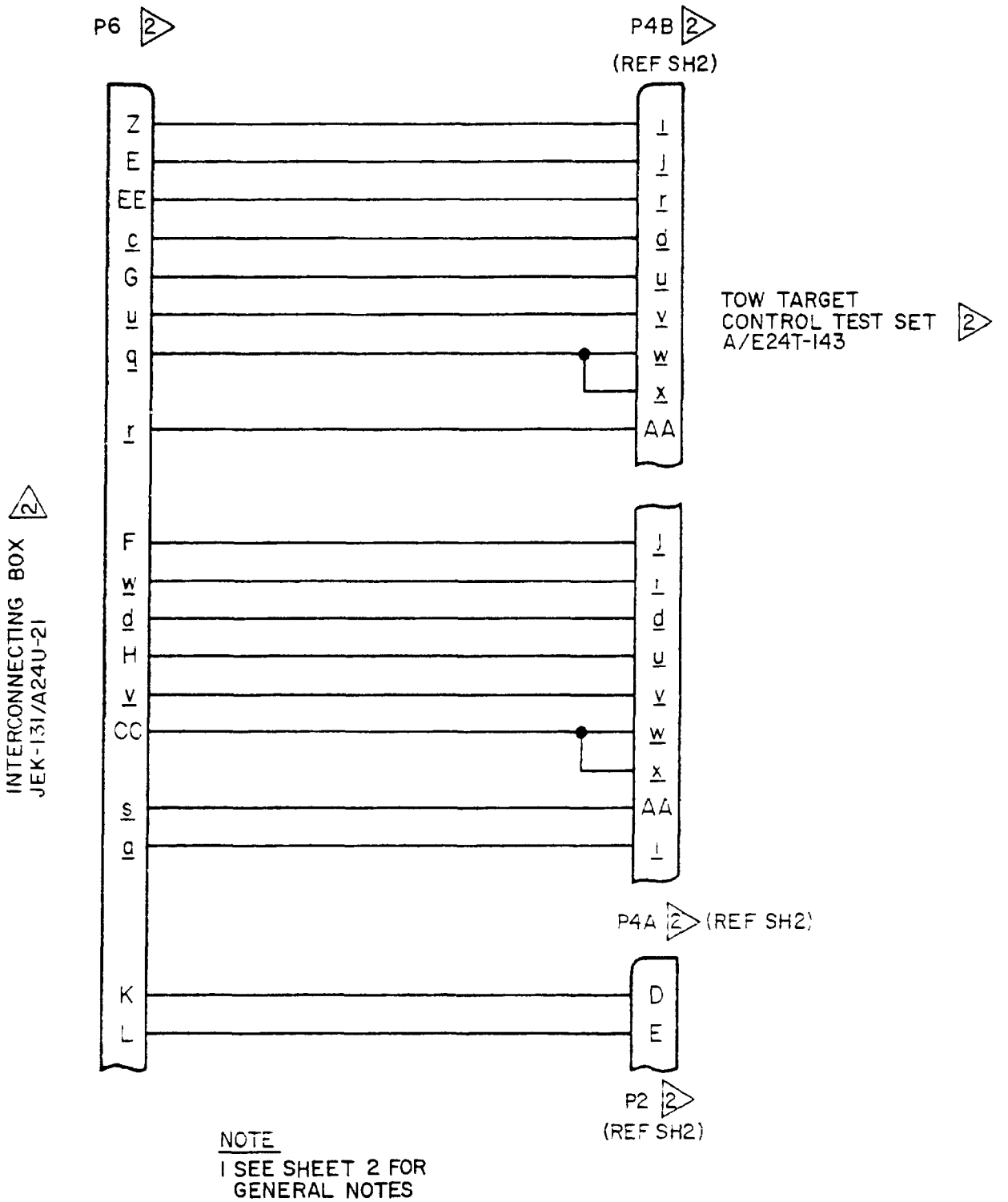
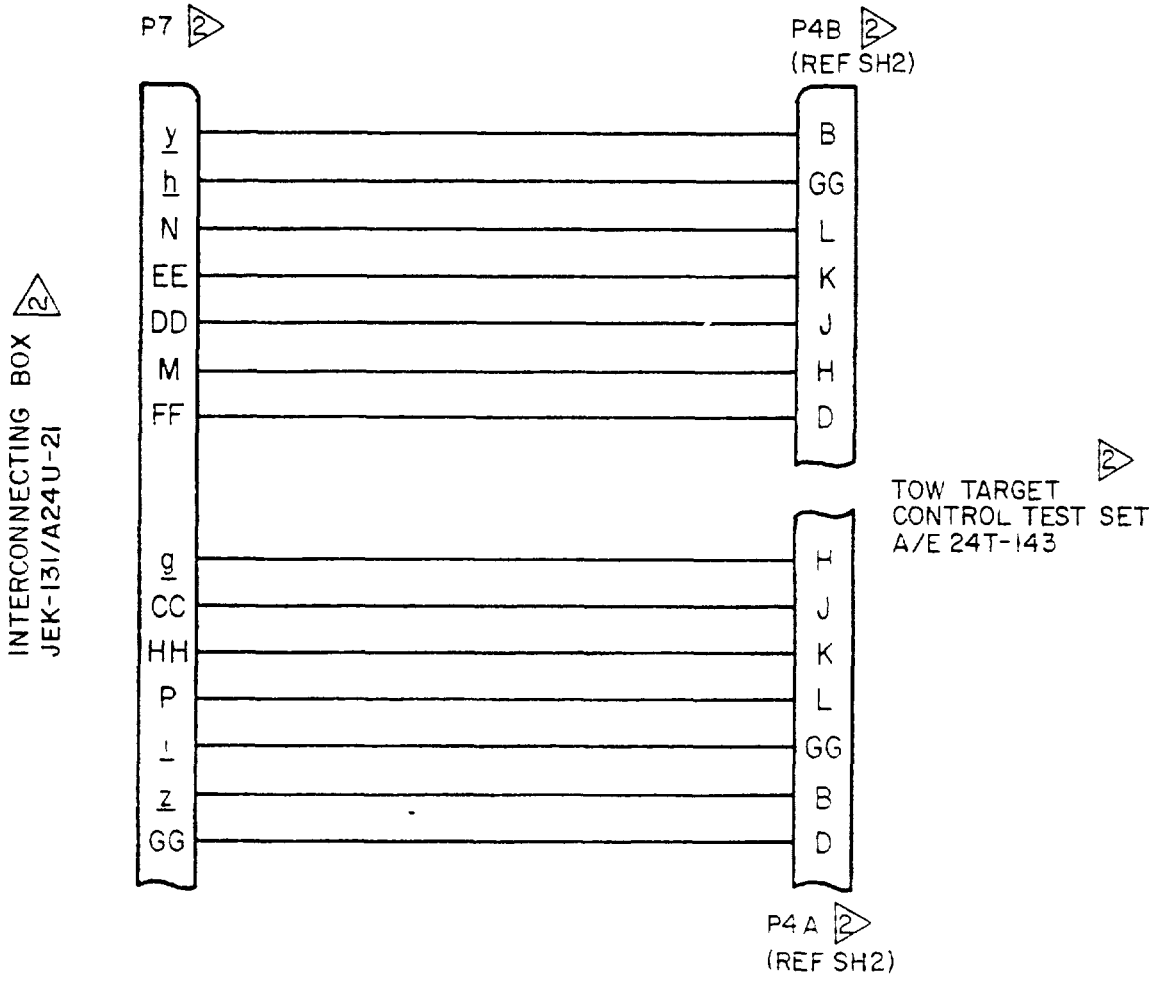


FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)

SHEET 6 OF 9

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NOTE
1 SEE SHEET 2 FOR GENERAL NOTES

FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)

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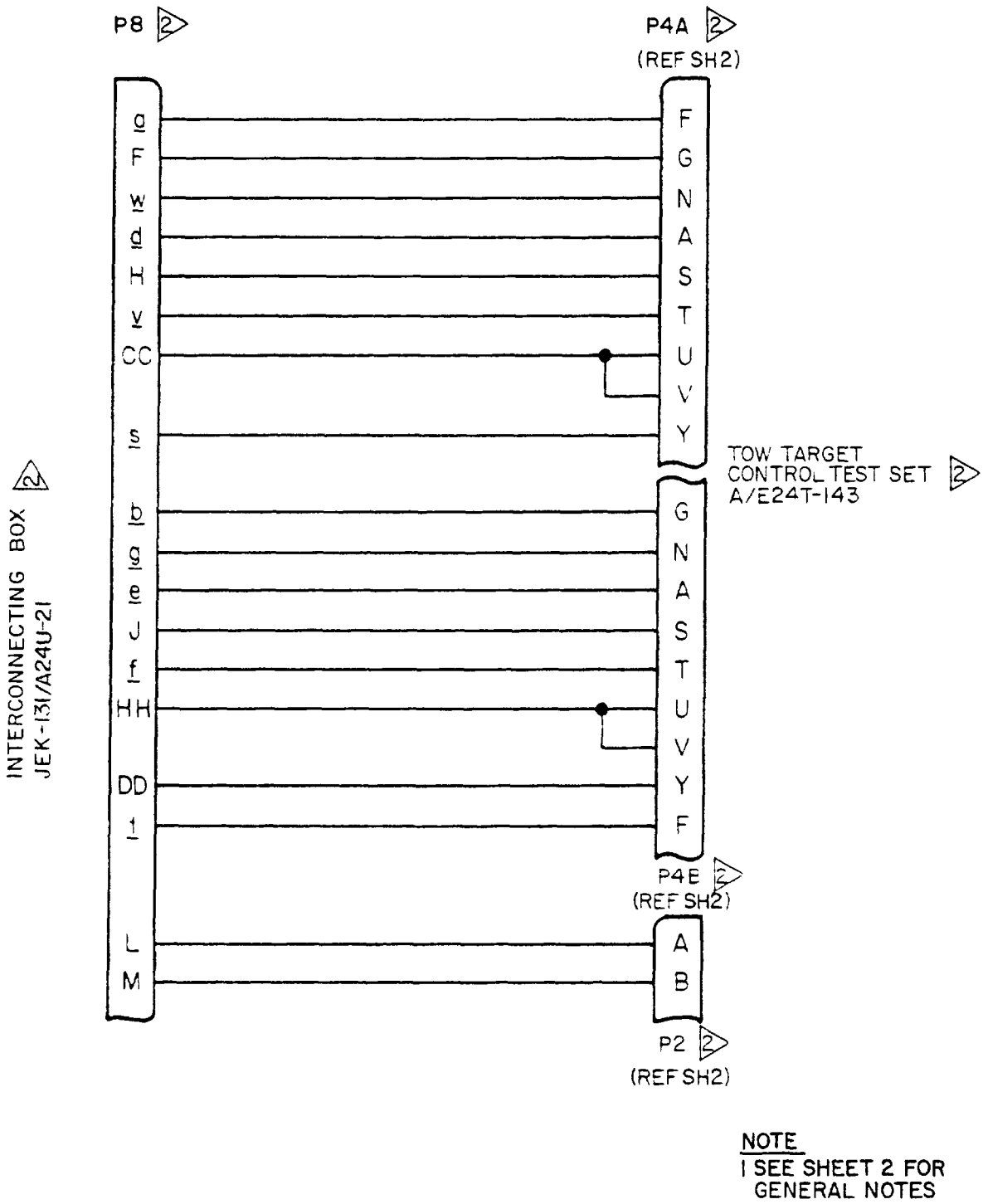


FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)

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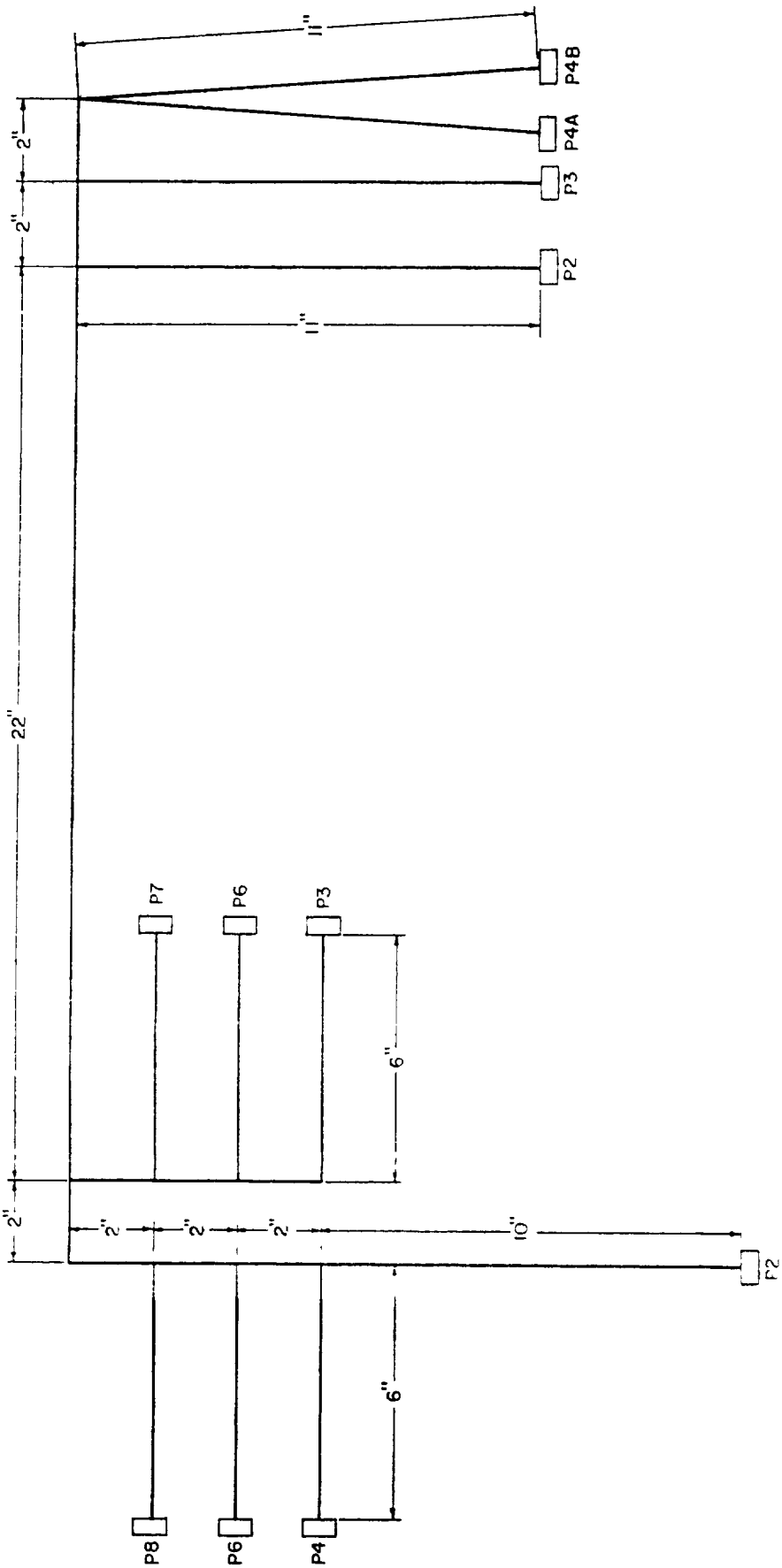


FIGURE 2. INTERCONNECTING BOX, BENCH TEST SET-UP. - (CONTINUED)
(BENCH TEST CABLE FOR FIGURE 1)

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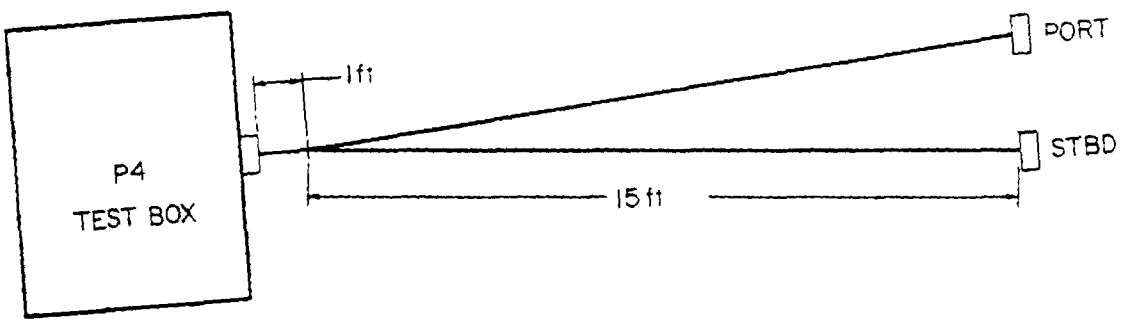


FIGURE 3. AIRCRAFT TEST CABLE, TOW TARGET SYSTEM.

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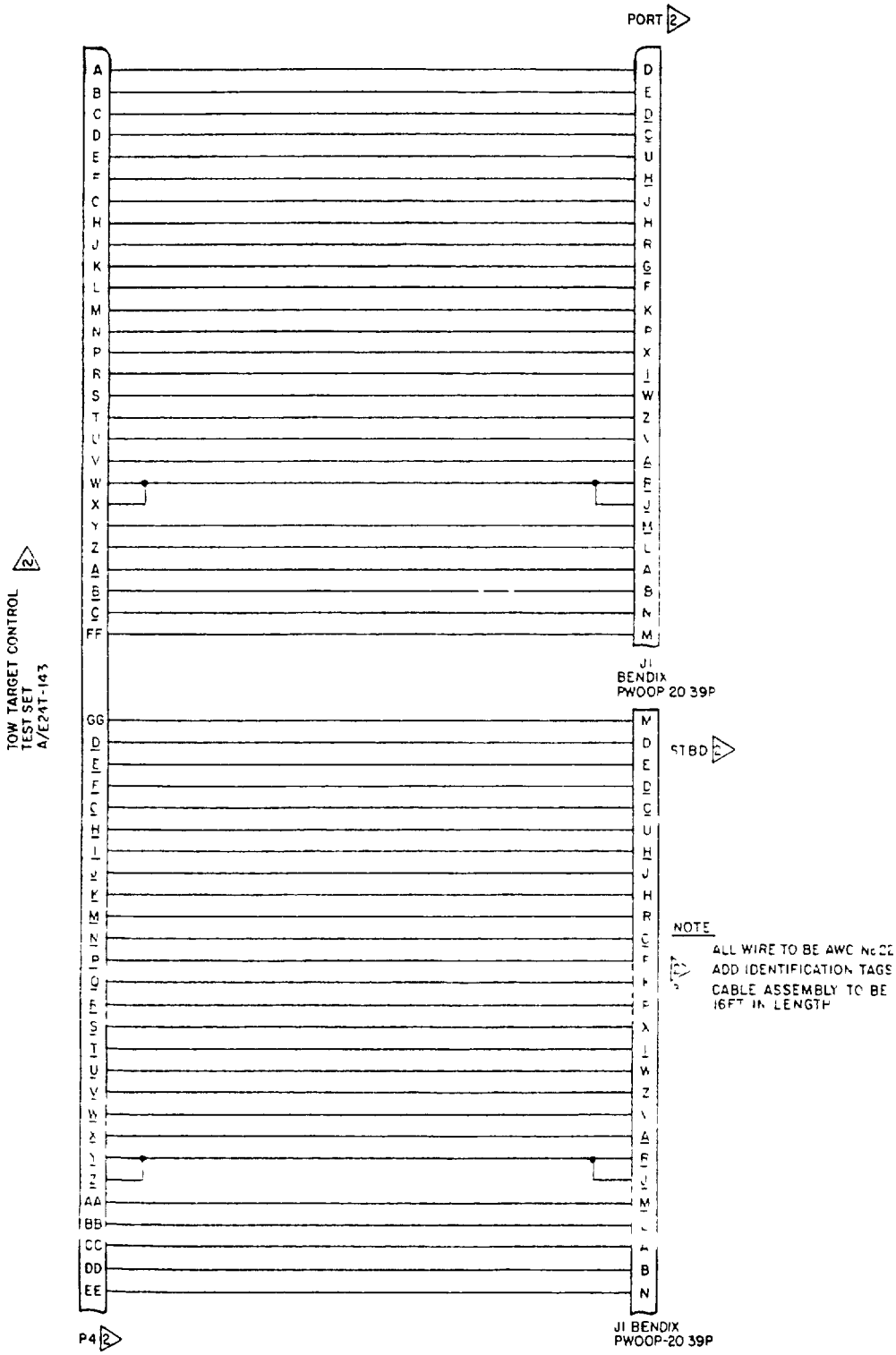


FIGURE 3. AIRCRAFT TEST CABLE, TOW TARGET SYSTEM.

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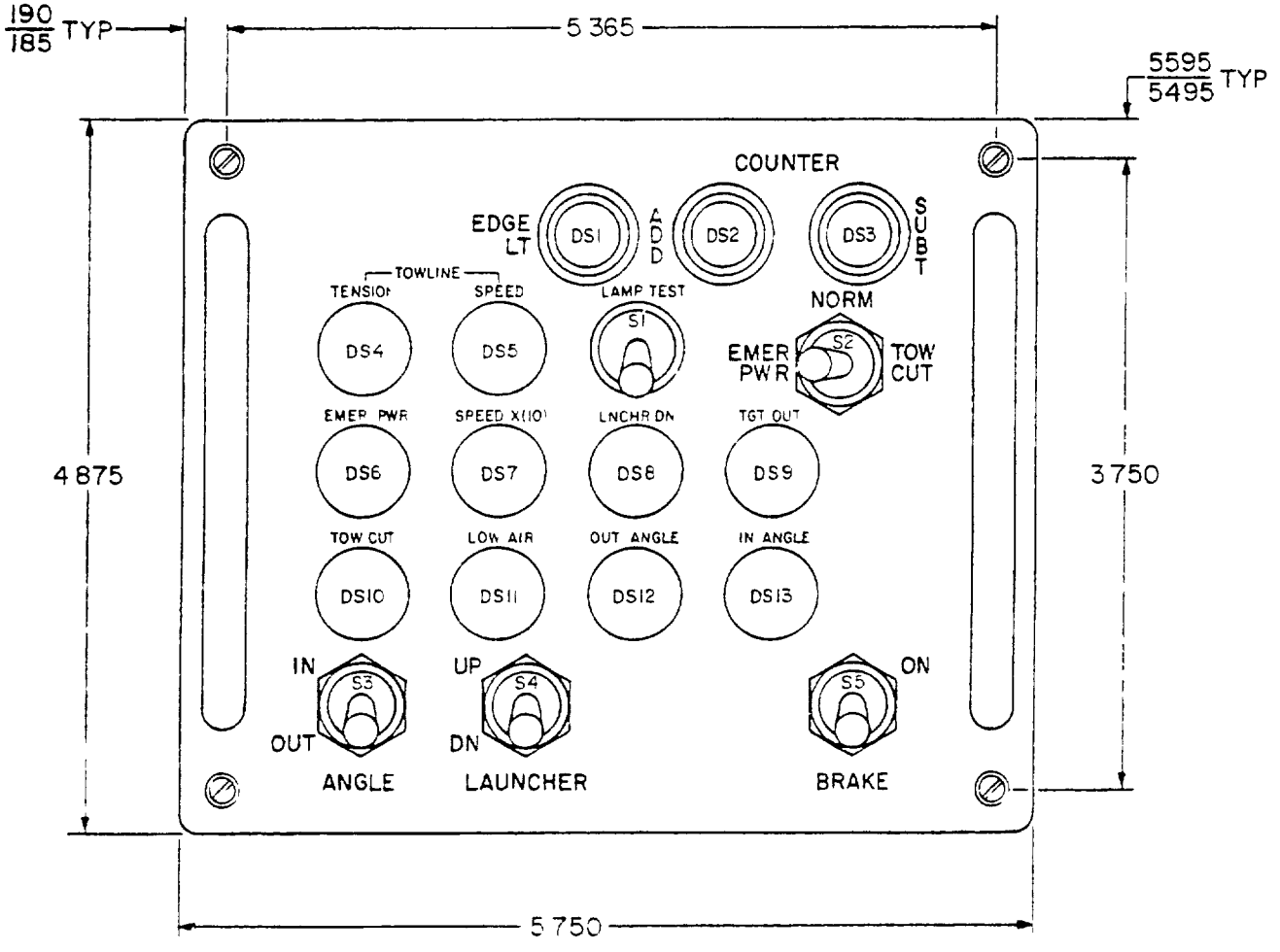
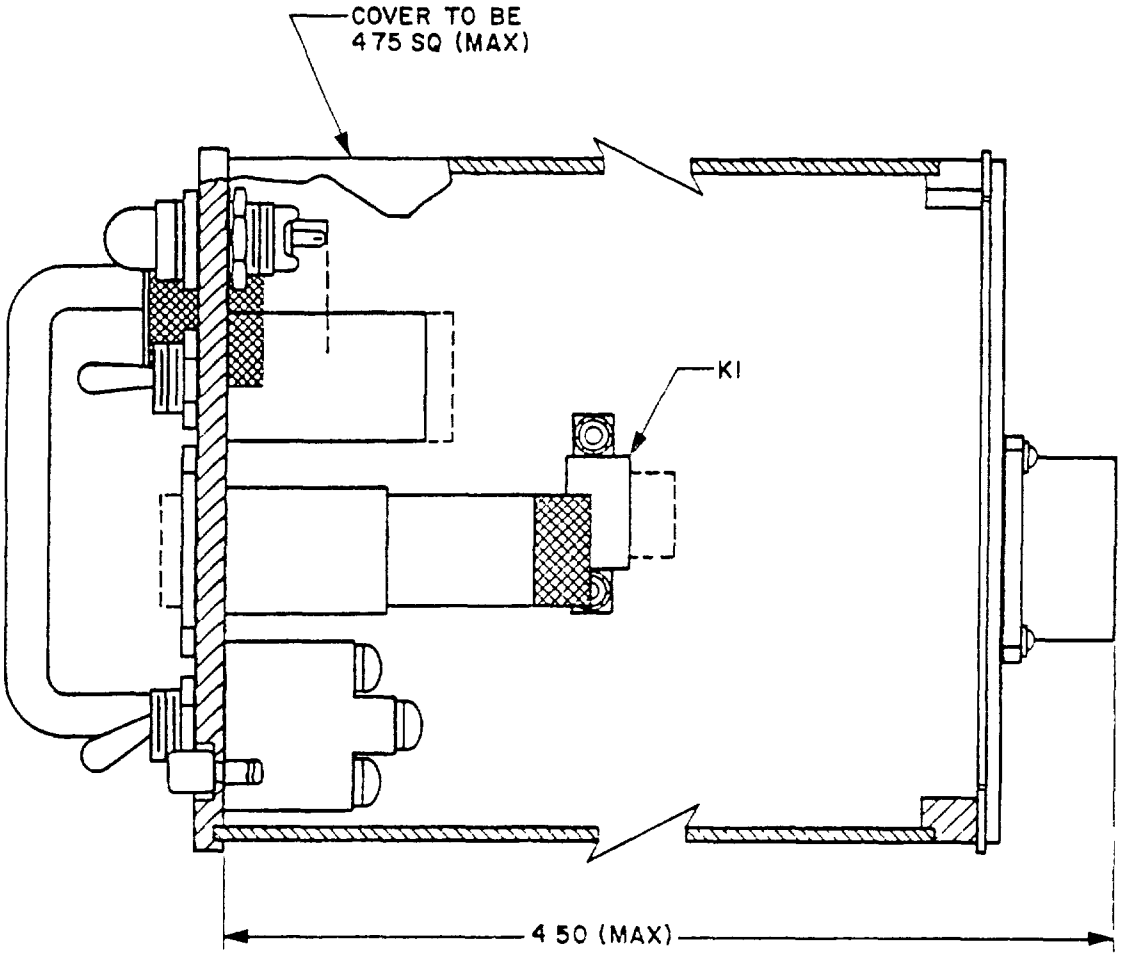


FIGURE 4. SUGGESTED FRONT PANEL LAYOUT, REEL LAUNCHER CONTROL SIMULATOR.

SHEET 1 OF 2

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SIDE VIEW

FIGURE 4. REEL LAUNCHER CONTROL SIMULATOR - SIDE VIEW.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-85530(AS) TEST SET, TOW TARGET, A/E24T-143

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

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